# On-Chain Prediction Market

- Upgradeable binary markets
- ERC20Votes governance + Timelock
- ERC1155 outcome shares and ERC4626 fee vault
- Foundry: 86 tests passing

# Problem

- Prediction markets need transparent market creation, reliable resolution, auditable payouts, and governance controls.
- The project demonstrates a protocol surface, not only one isolated contract.

# Architecture

- UUPS PredictionMarket proxy is the core.
- OracleAdapter wraps Chainlink-style feed checks.
- GovernanceToken, PredictionGovernor, and TimelockController control admin actions.

# Smart Contracts

- PredictionMarket: lifecycle, buying, resolving, claiming, fees
- MarketFactory: CREATE and CREATE2 deployment
- OutcomeToken: ERC1155 YES/NO token IDs
- FeeVault: ERC4626 collateral vault

# Governance Flow

- Tested lifecycle: propose -> vote -> queue -> timelock delay -> execute.
- The proposal executes MarketFactory.deployToken through TimelockController.
- Parameters: 1 day delay, 1 week period, 4% quorum, 2 day timelock.

# Security

- Owner-only privileged functions
- UUPS owner initialization fixed
- ReentrancyGuard on token-flow functions
- Checks-Effects-Interactions for accounting
- SafeERC20 transfers and oracle staleness checks

# Testing

- 86 tests passed, 0 failed, 0 skipped
- Unit, fuzz, invariant, and optional fork-safe tests
- ERC1155, ERC4626, and Governor lifecycle tests included.

# Frontend + Subgraph

- Frontend scaffold: wallet connect, reads, create market, delegate, vote, subgraph query.
- Subgraph entities: Market, SharePurchase, Resolution, FeeWithdrawal.

# Deployment

- Deploy.s.sol deploys the full protocol stack.
- PostDeployCheck.s.sol verifies Governor, Timelock, and factory ownership.
- Real L2 addresses require PRIVATE_KEY and RPC_URL in local .env.

# Defence Demo Plan

- Run forge build and forge test.
- Show PredictionMarket, PredictionGovernor, OutcomeToken, FeeVault.
- Explain remaining network-only deployment steps.